Evaluating the Trustworthiness of Explanations for Student Dropout Risk: Comparing Black-Box and Interpretable Models

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Abstract

University support offices can hold only a limited number of outreach meetings and must therefore decide early which students to contact. This study examines whether a tabular foundation model, TabICL, should be used to rank students for such meetings after the first semester, and whether its explanations are trustworthy and comprehensible. Using 3,630 Dropout or Graduate records from the public UCI Predict Students' Dropout and Academic Success *dataset, all modelling choices were fixed on a validation split before a single locked final test, in which each of five models flagged the top 20% of students, the assumed outreach capacity. TabICL achieved a ROC-AUC of 0.957 and flagged 109 test students, 108 of whom dropped out (precision 0.991, recall 0.507); however, it was not reliably better than XGBoost (difference 0.001, 95% interval* -0.006 *to 0.009) or an interpretable Explainable Boosting Machine (0.003,* -0.007 *to 0.014). A post-hoc audit of all 544 validation students showed that the SHAP explanations were faithful to the model but often unstable: when the comparison group changed, 411 students fell below the 0.80 stability threshold for their top five reasons. Moreover, the flags largely restated information already visible to advisers: 78 of the 108 flagged validation students had passed no first-semester units, and first-semester progress was the leading reason for every flagged student. In a 50-person study, there was no evidence that explanation charts improved participants' predictions of the model's flag (0.900 with charts versus 0.916 without; difference* -0.016*, 95% interval* -0.060 *to 0.032). On this dataset, these findings indicate that a simpler interpretable model matched the foundation model's predictive performance and is the better candidate for further validation, that explanations should be presented with their uncertainty, and that any use should support, not replace, an adviser's judgement.*

Keywords: student dropout risk, explainable artificial intelligence, tabular foundation models, interpretable machine learning, SHAP, human-centred evaluation

Summary for student-support staff. *The decision.* After the first semester, a support office can meet only about one in five students and must decide whom to contact first. *What the model does.* It assigns each student a dropout-risk score from records the university already holds (first-semester results, fees and background) and places the top 20% on a contact list. *Main findings.* On past data, the list had very high precision: 108 of the 109 flagged test students dropped out. However, on validation data most flagged students had passed no units at all, which staff can already see, and the list still misses about half of the students who later leave, partly because only one in five can be contacted. A simpler, fully transparent model had comparable final-test performance within the paired-bootstrap uncertainty. The "reasons" displayed are useful hints but may change when recalculated, and there was no evidence that seeing them helped people predict the model's decision. *Limitations.* The model was tested only on Portuguese data from students with a known final outcome (still-enrolled students were excluded). It does not explain *why* a student is struggling or what will help, and it must never be used to deny support.

1. Introduction

1.1. Background and Problem Statement

Student dropout is costly for students and institutions alike, and the dataset used here was created specifically to help identify at-risk students early enough for support (Realinho et al., 2021, 2022). In practice, a student-support office can hold only a limited number of outreach meetings after the first semester and must therefore rank students and contact those at highest risk first. Ranking alone, however, is insufficient: before a conversation, an adviser also needs to understand *why* a student was flagged, and those reasons must be trustworthy.

Tabular foundation models such as TabPFN (Hollmann et al., 2025) and TabICL (Qu et al., 2025) have been reported to match or outperform tuned gradient boosting on many small tabular benchmarks without dataset-specific training. They are, however, black boxes whose explanations must be obtained post hoc, for example with SHAP (Lundberg and Lee, 2017), and for high-stakes decisions an interpretable

model should arguably be preferred whenever it performs comparably (Rudin, 2019).

Problem statement. Benchmark accuracy does not establish that a foundation model improves a capacity-limited outreach decision, nor that its post-hoc explanations are faithful, stable or useful to the people who must act on them, and none of the representative studies in Table 2 combines these three perspectives for dropout outreach. This project therefore asks whether a foundation model's complexity is justified here and whether its explanations withstand technical and human evaluation.

1.2. Decision Setting

The intended user is a student-support officer rather than the student. Predictions are made once, after the first semester; second-semester information is excluded, as it would not yet be available. The operating rule was fixed in advance: all students in a cohort are ranked by predicted dropout probability, and exactly the top 20% (the assumed outreach capacity) are flagged, with ties broken by record order. A flag means "offer a support conversation"; it never implies withdrawing support or making an automated decision about the student. Errors carry different costs: a missed student may leave without being offered help, whereas an unnecessary flag uses a scarce meeting and may stigmatise a student who was coping, and a misread explanation may steer the conversation towards the wrong problem.

1.3. Research Questions

RQ1. Does TabICL add predictive value over simpler conventional and intrinsically interpretable baselines at a fixed 20% outreach capacity?

RQ2. Are its local explanations faithful, stable and appropriately constrained, and what do they reveal to an adviser about the students it flags?

RQ3. Do explanation charts help people anticipate the model's outreach flag and identify useful support information?

1.4. Contributions

1. A pre-specified, locked comparison of TabICL with logistic regression, a depth-3 decision tree, XGBoost and a main-effects EBM at the same outreach capacity, with paired bootstrap intervals **(RQ1)**.
2. A post-hoc audit of the explanations for all 544 validation students, covering faithfulness, stability and constrained recourse, together with an analysis of which students are flagged **(RQ2)**.
3. A two-group forward-simulation study with 50 adult participants, including a qualitative analysis of 1,500 written answers **(RQ3)**.
4. Sensitivity analyses for still-enrolled students, an admission-time feature window and a model without protected attributes, together with a local, validation-only research interface that illustrates how the outputs could be presented to staff.

1.5. Scope of the Work

The study relies on one public dataset from a single Portuguese institution, a binary Dropout-versus-Graduate target and one fixed train/validation/test split, and its fairness analysis is descriptive. The human study involved volunteers acting as proxy users rather than support officers; it therefore measures understanding of the model's output, not decisions about real students or their outcomes. Table 1 defines the key terms.

Table 1. Key terms used in this report.

Term	Meaning
Outreach capacity	Share of students the office can meet: here the top 20% of a cohort
Flag	The student is in that top 20% and would be offered a meeting
Risk score	The model's estimated dropout probability; a score, not a certainty
Contribution (SHAP)	How far one group of information moved a student's score relative to a typical student; association, not cause
Faithfulness	Whether the reasons shown are those the model actually relies on
Stability	Whether the same top five reasons reappear when the explanation is recalculated
Recourse	A change to the record that would remove the flag; not advice
Forward simulation	People predict the model's decision from the same information
Locked final test	Held-out data used exactly once, after all choices were fixed

2. Related Work

2.1. Student Dropout Prediction

Realinho et al. (2022) released the dataset used in this study (Realinho et al., 2021): 4,424 undergraduates from a Portuguese polytechnic, described by demographic, socio-economic, application, first- and second-semester academic and macro-economic attributes and labelled Dropout, Enrolled or Graduate, and reported three-class prediction with standard classifiers. The present study instead fixes a single prediction time, excludes later-semester information and evaluates at an explicit outreach capacity, which are the conditions under which a support office would use the scores.

2.2. Tabular Foundation Models and Interpretable Baselines

TabPFN (Hollmann et al., 2025) and TabICL (Qu et al., 2025) are transformers pretrained on synthetic tables that predict a new row by reading the training rows as context, without gradient updates on the target data; TabICL extends this approach to larger tables through column-then-row attention.

Table 2. Representative prior work and the gap this project addresses.

Study	Focus	Method	Gap for this project
Realinho et al. (2022)	Dropout dataset (UCI)	Dataset description, ML baselines	Three-class accuracy; no capacity rule, explanation audit or users
Qu et al. (2025); Hollmann et al. (2025)	Tabular foundation models	In-context learning on tables	Benchmarks accuracy; not explanation quality or decision use
Lou et al. (2013); Caruana et al. (2015)	Interpretable models	Generalised additive models (EBM)	Glass-box alternative; predates, and is not compared with, foundation models
Lundberg and Lee (2017)	Local explanations	Shapley-value attributions	Approximations depend on budget and background data
Tiukhova et al. (2024)	Learning analytics	Stability of XAI for student success	Stability of explanations, but no human evaluation
Nauta et al. (2023)	XAI evaluation	Systematic review (Co-12 properties)	Calls for quantitative, multi-property evaluation
Hase and Bansal (2020)	Human evaluation	Forward simulation	Not applied to dropout outreach
Karimi et al. (2021)	Recourse	Counterfactuals vs. interventions	Feasible changes need causal care
This project	Dropout outreach at 20% capacity	TabICL vs. 4 baselines, SHAP audit, 50-person study	Combines predictive, technical-explanation and human evidence

Strong conventional baselines include logistic regression and gradient-boosted trees such as XGBoost (Chen and Guestrin, 2016). Explainable Boosting Machines (EBMs), by contrast, are boosted generalised additive models whose per-feature shape functions constitute the explanation (Caruana et al., 2015; Lou et al., 2013). Rudin (2019) argues that such models should be preferred in high-stakes settings when accuracy is comparable, a claim this study tests directly.

2.3. Evaluating Explanations

SHAP attributes a prediction to features using Shapley values (Lundberg and Lee, 2017). For black-box models these values are estimated by sampling and therefore depend on the sampling budget, the random seed and the background data that defines the "typical" reference. Nauta et al. (2023) recommend measuring properties such as correctness (faithfulness) and consistency (stability) quantitatively rather than relying on anecdotes, and Tiukhova et al. (2024) show that explanations of student-success models can be unstable across refits. Counterfactual explanations (Wachter et al., 2017) describe which change would alter a prediction; however, Karimi et al. (2021) caution that a feasible change to a feature is not equivalent to an effective real-world intervention. Global behaviour can be summarised with accumulated local effects (ALE) (Apley and Zhu, 2020).

2.4. Human-Centred Explainability Evaluation

Doshi-Velez and Kim (2018) distinguish application-grounded, human-grounded and functionally-grounded evaluation. Forward simulation, in which people predict a model's output from its input with or without an explanation, is a common human-grounded test (Hase and Bansal, 2020). It measures whether an explanation makes the model's behaviour predictable, not whether people trust the model appropriately or make better decisions; this distinction is maintained throughout the interpretation of the study.

3. Data and Study Design

3.1. Dataset and Target

The study uses the UCI *Predict Students' Dropout and Academic Success* dataset (repository dataset 697, CC BY 4.0) (Realinho et al., 2021, 2022), which describes 4,424 students from a Portuguese higher-education institution with 36 predictors and three outcomes: Dropout (1,421), Enrolled (794) and Graduate (2,209). The distributed file contains no direct identifiers, missing values or duplicate rows. The task is binary, Dropout (1) versus Graduate (0), yielding 3,630 records. Enrolled students were excluded since their final outcome is unknown. This exclusion is a genuine limitation: at the moment of decision every student is enrolled, so a deployed model would also score students resembling those removed. Section 5.5 examines how the model treats them.

Eligibility. Of the dropouts, 17.3% (246 of 1,421) took no first-semester evaluations, compared with only 3.4% of graduates (75 of 2,209). Some labelled dropouts may therefore have left before the decision point rather than after it, in which case not every flagged "dropout" could still have been reached by a support meeting. This timing ambiguity is reported as an explicit eligibility limitation; Section 5.2 analyses who is flagged, and Section 5.5 re-ranks students after setting these cases aside.

Exploratory data analysis. Table 3 compares dropouts and graduates on the training split only (2,541 students), leaving the test split uninspected. The classes are moderately imbalanced (39% dropout). First-semester results separate the groups most clearly: dropouts passed 2.6 units on average against 6.3 for graduates, and 40% passed none. Financial indicators also differ (fees up to date 69% vs. 99%; debtor 22% vs. 5%), as do age and gender. These patterns anticipate both the model's reliance on first-semester progress and the subgroup gaps reported in Section 5.3.

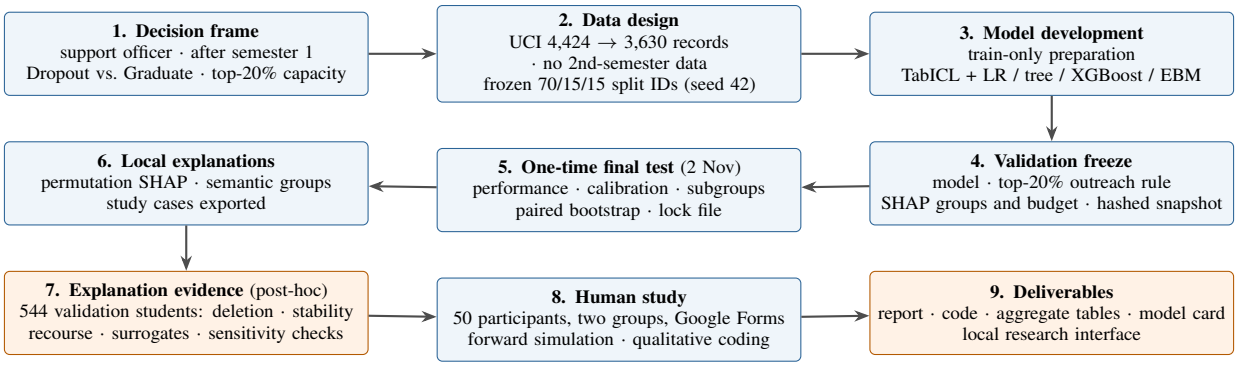


Figure 1. Research workflow. The model, outreach rule and explanation settings were frozen before the one-time final test; orange steps were added afterwards without changing the model.

Table 3. Training-split profile by outcome (2,541 students; grade 0–20, and 0 if no unit was passed).

Characteristic	Dropout	Graduate
Students	995	1,546
First-semester units passed (mean)	2.6	6.3
Passed no first-semester unit	39.6%	3.2%
Took no first-semester evaluation	17.6%	3.0%
First-semester average grade (mean)	7.3	12.7
Tuition fees up to date	68.7%	98.6%
Debtor	21.5%	5.0%
Scholarship holder	9.8%	39.1%
Age at enrolment (median)	23	19
Female (gender 0)	52.4%	77.0%

3.2. Prediction Time and Feature Windows

The primary feature window is *after the first semester*. It contains 30 original attributes covering background (age, gender, nationality, marital status, displacement and special needs), application (route, order, course, attendance, prior qualification and grades), family (parents' education and occupation), finances (scholarship, tuition fees up to date and debtor status), first-semester units (credited, enrolled, evaluated, passed, without evaluation and average grade) and national economic indicators. Three derived features complete the 33 inputs: the first-semester pass rate and indicators for no units enrolled and no evaluations taken. All six second-semester attributes are removed to prevent leakage of future information, and an admission-time window without first-semester results serves as a sensitivity analysis.

3.3. Frozen Split and Outreach Policy

The 3,630 records were split once, with stratification, into 70/15/15% train, validation and test sets (2,541, 544 and 545 records; seed 42), and the row identifiers were saved. Every modelling and explanation setting was chosen on the validation split. The test split was subsequently used exactly once, in a run protected by a lock file and a hash of the frozen code, data and model.

The outreach policy flags the $k = \lfloor 0.2n \rfloor$ highest-scoring students in a cohort of size n (108 of 544 in validation and 109 of 545 in test), breaking ties by record order. Consequently, there is no universal probability threshold: a student's flag depends on their rank within the cohort.

3.4. Human-Study Design and Ethics

The study applies forward simulation (Hase and Bansal, 2020), asking whether people can predict the model's hidden flag from a student profile. Fifty adult volunteers (a convenience sample, with about ten recruited by each team member) were assigned by a sealed, seeded list to Group A (profile only, $n = 25$) or Group B (profile plus a bar chart of the five largest grouped SHAP contributions, $n = 25$). The chart displayed direction and relative size only, never the score, the base value or the flag (Figure 2).

Each participant first completed one unscored practice case from the validation split and then ten scored test cases (five flagged and five not flagged by the model) in one of two fixed orders, a seeded order or its exact reverse, yielding four Google Forms. For each case, participants predicted whether the model had flagged the student, rated their confidence from 1 to 5 and answered three short written questions: why the model decided as it did, what support they would suggest and what further information they would want. Group B additionally answered two chart-reading checks on the practice case, and all participants self-reported their total completion time.

Ethics and data governance. As a class project, the study followed basic research-ethics practice (voluntary informed consent, minimal data, no identifiers and restricted access) rather than formal ethics-board review. All participants were adults, and taking part or declining had no effect on grades, employment or access to services. No names or AIT student records were collected; responses carry random participant codes, the case lookup is stored separately, raw free text is held outside version control with access limited to the team, and only aggregate results are reported. Participants could withdraw through their code until their responses

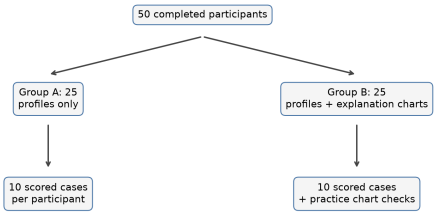


Figure 2. Two-group forward-simulation design (25 participants per group).

entered the aggregate analysis; all 50 completed the study and none were excluded. Raw responses and case-linkage files remain in access-restricted storage, excluded from the public repository and used only for this course project, and the written answers were never used to train, validate or explain the model. The online, unsupervised format is discussed as a limitation.

4. Methods

4.1. Predictive Models

Table 4 lists the five models. All receive the same 33 inputs, with coded administrative attributes (for example course, application route and parents' occupations) treated as categorical. Logistic regression, the decision tree and XGBoost share one preprocessing step (one-hot encoded categoricals; median-imputed, standardised numericals), whereas EBM and TabICL receive the named columns directly. All settings were fixed in advance in a written analysis plan rather than tuned on the test set.

Model complexity. The models differ in how far a person can follow them. TabICL is a 27.6-million-parameter transformer with 18 attention blocks (3 column, 3 row and 12 in-context), which scores each student by attending over all 2,541 training students as context and averages 4 ensemble members. It is therefore neither simulatable nor decomposable, since its reasoning cannot be recovered from a coefficient list or a small tree, and it requires post-hoc explanation. The main-effects EBM, in contrast, is interpretable by construction: each prediction is exactly the logistic transform of an intercept plus one learned shape-function contribution per feature (its validation probabilities are reconstructed from these terms within 3.3×10^{-16}). Logistic regression and the depth-3 tree are also simulatable.

Validation-stage selection. On the validation split, ROC-AUC was 0.952 for TabICL, 0.946 for EBM, 0.945 for logistic regression and XGBoost, and 0.923 for the tree. As the margin of TabICL over EBM (0.006) was modest, the proposal had already identified EBM as a potentially more defensible candidate and deferred the decision to the final test.

Table 4. Models compared at the same 20% outreach capacity.

Model	Configuration
Logistic regression	One-hot categoricals, balanced class weights
Decision tree	Depth 3, balanced class weights
XGBoost	300 trees, depth 4, learning rate 0.05, subsampling 0.8 (Chen and Guestrin, 2016)
EBM	Main effects only (no interactions), interpret 0.7.8 (Lou et al., 2013)
TabICL	v2.2.0, 4 ensemble members, CPU, official checkpoint (Qu et al., 2025)

4.2. Predictive and Capacity-Based Evaluation

Ranking quality is measured by ROC-AUC, and probability quality by the Brier score and the 10-bin expected calibration error (ECE). Since the decision is to "flag the top 20%", the principal operating metrics are dropout recall and precision among flagged students:

$$\text{Recall@20\%} = \frac{\#\{\text{flagged dropouts}\}}{\#\{\text{dropouts}\}}, \tag{1}$$

$$\text{Precision@20\%} = \frac{\#\{\text{flagged dropouts}\}}{\#\{\text{flagged}\}}. \tag{2}$$

With a fixed capacity, recall is capped: with 213 dropouts and 109 slots in the test set, at most 51% of dropouts can be flagged. Uncertainty in the differences between TabICL and each baseline is estimated with 2,000 paired bootstrap resamples of the test set (seed 42).

4.3. Local and Global Explanations

Local explanations are computed with permutation SHAP (Lundberg and Lee, 2017) on the predicted dropout probability, using 25 training rows as the background (the "typical student" reference), kept small since every SHAP evaluation scores the student against each background row. Signed SHAP values of related features are summed into semantic groups, such as *first-semester progress* (units passed, grade and pass rate) and *first-semester participation* (units enrolled and evaluated), and the five groups with the largest absolute values are displayed. The SHAP budget is $\texttt{max_evals} = 67 = 2 \times 33 + 1$ model evaluations per student, which is exactly one forward-and-backward pass over one random feature ordering, the smallest budget shap's permutation explainer accepts. Exact TreeSHAP is unavailable, as TabICL is not a tree ensemble. Before the final test, the pipeline was verified: SHAP values reconstructed TabICL's probabilities within 4.1×10^{-8}, and on a synthetic white-box model the explainer recovered the known feature ranking exactly (Spearman 1.0). Global behaviour is summarised by grouped permutation importance and by ALE (Apley and Zhu, 2020), which is less affected than partial dependence by correlated features.

4.4. Explanation Audit

The analysis plan originally scheduled the full audit on 20 validation students. Owing to the project timeline, it was instead conducted after the final test on all 544 validation students, using a TabICL refit with the frozen settings. This refit reproduces the frozen model on validation (largest probability difference 1.0×10^{-5}; all 108 flags identical); it therefore describes the tested model, although it constitutes post-hoc evidence rather than a gate passed before the final test.

- **Faithfulness.** Groups are replaced by training medians or modes in SHAP order and in 50 random orders; a positive area between the two cumulative probability-change curves indicates that the SHAP ranking identifies the groups on which the model relies.
- **Stability.** Each explanation is recomputed 10 times (45 pairwise comparisons per student, a workload still feasible for all 544 students) under three variations: a different random seed, a different 25-row background sample and minimal grade perturbations (Gaussian noise of 1% of the training standard deviation, clipped to the training range and accepted only if the prediction moves by at most 0.01). Stability is the mean pairwise overlap of the top-five groups, since five bars are what an adviser sees; values below 0.80, meaning that more than one of the five reasons typically changes, are flagged as unstable. All 544 students are used so that instability can be counted rather than illustrated.
- **Constrained recourse.** Features were classified in advance by availability and feasibility: age, nationality, prior qualifications and completed first-semester results remain fixed. For flagged students, only the administrative fields *debtor* and *tuition fees up to date* may change, as these could realistically be resolved, often with institutional help, before the second semester. The smallest such change is deemed to work if the student leaves the top 20% when re-ranked with all other scores fixed; students without a working change are reported explicitly.
- **Surrogates and pairs.** Six local linear surrogates (OLS, ridge and lasso, each with and without pair terms) and pairwise group masking assess how additive the model's local behaviour is.

4.5. Human-Study Analysis

The pre-specified primary outcome is each participant's accuracy over the ten cases, defined as agreement with the model's flag. Group means are reported with 95% intervals from 10,000 stratified bootstrap resamples, and the difference between Group B and Group A with its interval, a two-sided permutation p-value (10,000 label permutations) and Hedges' g. With 25 participants per group, these comparisons are exploratory. Confidence and self-reported time are analysed in the same way, while case-level accuracy, a leave-one-case-out sensitivity analysis and the chart-reading checks are reported descriptively.

4.6. Qualitative Coding

The 1,500 written answers (50 participants $\times$ 10 cases $\times$ 3 questions) were coded with a pre-written codebook, assigning one primary theme per answer before group membership was attached. Each support suggestion was additionally rated on four items from the proposal: whether it names a concrete next step, whether it could begin before the second semester, who is responsible (the student, the institution or both) and whether it asserts an unsupported cause. The planned independent double-coding of 20% of answers was not carried out; inter-coder agreement was therefore not measured, and the themes represent a single descriptive coding pass.

5. Results

Unless stated otherwise, predictive results come from the single locked final test (545 students, 213 dropouts), whereas the explanation-audit and sensitivity results use only the 544 validation students and are post-hoc.

5.1. Predictive Performance on the Locked Final Test

Table 5 and Figure 3 summarise the five models. TabICL achieved the highest point ROC-AUC (0.957), although XGBoost (0.956), EBM (0.954) and logistic regression (0.953) were all within 0.004. At the fixed capacity, the five models flagged between 107 and 109 of the 213 dropouts. TabICL flagged 109 students, of whom 108 dropped out (precision 0.991), and identified 108 of the 213 dropouts (recall 0.507); its confusion matrix was TN = 331, FP = 1, FN = 105 and TP = 108. XGBoost achieved the best Brier score and calibration, while EBM achieved the best capacity recall and precision.

Table 5. Locked final-test results (545 students, top 20% flagged; best in bold).

Model	AUC	Brier	ECE	Rec.@20%	Prec.@20%
Logistic reg.	0.9525	0.0822	0.066	0.507	0.991
Decision tree	0.9099	0.0999	0.069	0.502	0.982
XGBoost	0.9557	**0.0711**	**0.029**	0.507	0.991
EBM	0.9539	0.0768	0.040	**0.512**	**1.000**
TabICL	**0.9566**	0.0740	0.032	0.507	0.991

The paired bootstrap intervals (Table 6) show that TabICL clearly outperformed the depth-3 tree but could not be distinguished from XGBoost, EBM or logistic regression, as every interval for these comparisons includes zero. The answer to RQ1 is therefore that TabICL adds no demonstrated predictive value over simpler models on this task.

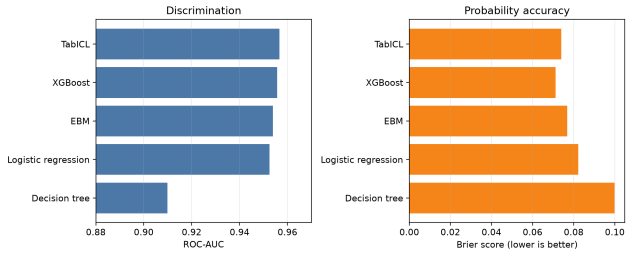


Figure 3. Final-test ROC-AUC and Brier score for the five models.

Table 6. TabICL minus each baseline on the final test (95% paired-bootstrap intervals; negative Δ Brier favours TabICL).

Comparison	Δ ROC-AUC	Δ Brier
vs. logistic reg.	0.004 [−0.005, 0.014]	−0.008 [−0.017, 0.001]
vs. decision tree	0.047 [0.029, 0.067]	−0.026 [−0.036, −0.016]
vs. XGBoost	0.001 [−0.006, 0.009]	0.003 [−0.003, 0.009]
vs. EBM	0.003 [−0.007, 0.014]	−0.003 [−0.010, 0.004]

5.2. Why Precision Is Nearly Perfect: Who Gets Flagged

A precision of 0.99 to 1.00 warrants scrutiny rather than celebration. On the validation split, TabICL flagged 108 students, all of whom dropped out, and Table 7 characterises them: 78 passed no first-semester units and 25 took no evaluations at all, while 19 of the 30 who passed at least one unit had unpaid tuition fees. A one-line rule, "flag students who passed zero units", selects 103 students with a precision of 0.874 and a recall of 0.423, compared with 1.000 and 0.507 for the model. On validation, the model's added value is therefore narrow: it excludes 25 zero-pass students (13 of whom graduated) and adds 30 students with some passed units, most of whom had fee problems. For most flagged students, therefore, an adviser would learn little beyond what the transcript already shows, and the flags arrive when these students may already have disengaged.

Table 7. The 108 flagged validation students compared with a simple rule.

Measure	Value
Flagged students who dropped out	108 / 108
Passed zero first-semester units	78
Took no evaluations at all	25
Passed ≥1 unit, tuition fees unpaid	19 of 30
Top explanation is first-semester progress	108 / 108
Rule "passed zero units": selected	103
precision / recall	0.874 / 0.423
TabICL at 20%: precision / recall	1.000 / 0.507

5.3. Calibration and Subgroup Diagnostics

Calibration error was small for all models except logistic regression and the decision tree (Table 5). Subgroup results on the final test are descriptive, and metrics are suppressed for groups of fewer than 30 students (all non-Portuguese nationalities, international students and students with special educational needs). Flag rates differed by gender (0.122 for group 0, female in the UCI coding, vs. 0.324 for group 1, male), displacement (0.136 for displaced vs. 0.288 for non-displaced students) and age (0.116 for students aged 20 or younger vs. 0.434 for ages 26–35). These differences largely follow the dropout base rates (0.30 vs. 0.54 by gender); however, recall also differed, at 0.410 vs. 0.593 by gender, 0.414 vs. 0.596 by displacement and 0.411 vs. 0.655 between the youngest and the 26–35 age bands. The model thus identifies a smaller share of dropouts among younger, female and displaced students (Appendix Table 11). Furthermore, several non-protected features act as strong proxies for protected attributes: application route is associated with international status (Cramér's $V = 0.61$) and age (0.50), and course with gender (0.40). The model's own leading inputs carry the same signal more weakly: on the final test, first-semester pass rate is associated with gender ($\eta = 0.28$) and age (0.24), and fee status with age ($V = 0.23$). Parental and financial fields may also encode socio-economic disadvantage. Removing protected columns therefore cannot remove their information.

5.4. Explanation Faithfulness and Stability

Faithfulness. Removing groups in SHAP order changed the prediction far more than removing them in random order: the mean area between the curves was 1.557 (95% interval 1.476–1.643) and was positive for 536 of 544 students (Appendix Figure 8). The explanations therefore point to information the model actually uses.

Content of the explanations. First-semester progress dominated (mean absolute grouped SHAP 0.277, in the top five for 97% of students), followed by participation (0.053), tuition fees (0.051) and course (0.042) (Figure 5). As first-semester progress was the leading reason for every flagged student, the explanation largely restates the transcript precisely for the students who matter most. Figure 4 illustrates how such an explanation could reach a support officer: a flagged validation student who sat ten evaluations but passed no unit, whose progress bar outweighs every other reason. Global methods agree (Appendix A.3).

Stability. The ranking of reasons was robust to minimal grade changes (mean top-five overlap 0.996; no student below 0.80) but not to the explainer's own randomness: overlap was 0.818 across random seeds (213 students below 0.80) and 0.745 across background samples (411 students below 0.80) (Figure 6).

Recourse and additivity. Only 17 of the 108 flagged students would leave the top 20% if their debt or fee status changed; for the remaining 91, no such administrative change exists. For example, for validation student 174, recording the fees as paid and the debt as cleared lowers the estimate from 0.996 to 0.689 and the cohort rank from

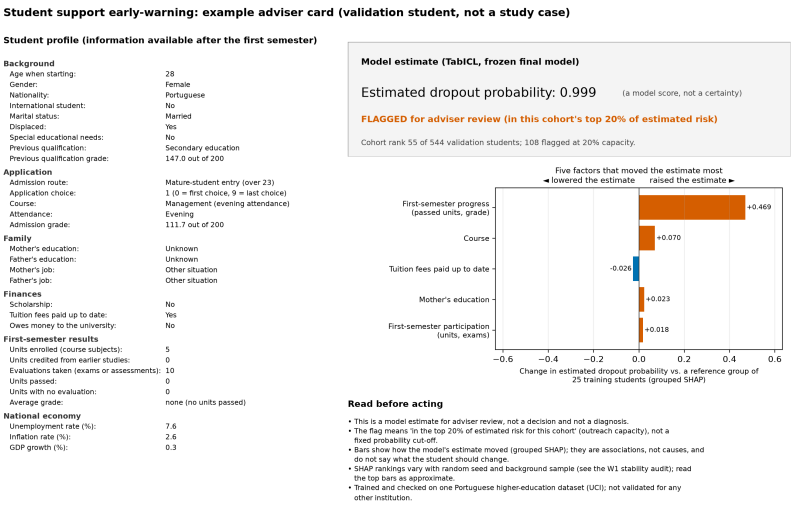


Figure 4. Example adviser card for a validation student: the record (left) and the flag, rank and five largest grouped SHAP reasons (right).

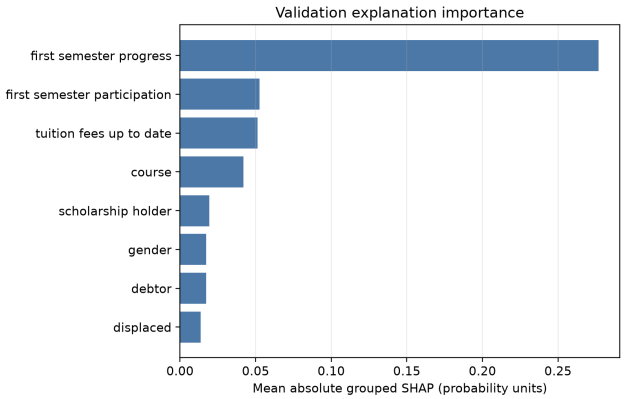


Figure 5. Mean absolute grouped SHAP contribution (544 validation students).

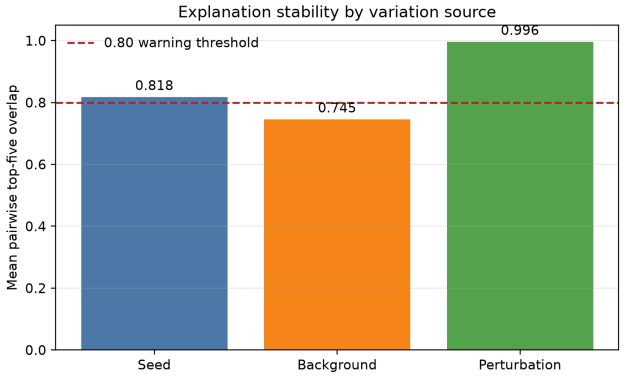


Figure 6. Top-five stability over 10 recomputations (dashed line: 0.80 warning level).

87 to 182, outside the 108 slots (Appendix A.7). The best local surrogate (lasso with pair terms) approximated TabICL with a mean absolute error of 0.022 on held-out masking patterns, against 0.073 without pair terms: the model's local behaviour is captured well only once pairwise interactions are included.

Summing the bars. SHAP contributions sum by construction to the prediction minus the comparison-group average, so all grouped contributions reproduce the ranking exactly. Even the five bars shown to participants separate flagged from unflagged students (AUC 0.97; 89 of 108 flags recovered).

5.5. Sensitivity Analyses

Enrolled students. The frozen model scores still-enrolled students between the two known groups (median 0.354, compared with 0.064 for graduates and 0.990 for dropouts). When a validation share of 119 enrolled students joins the cohort, they receive 7 of 132 flags (5%) despite forming 18% of the cohort; with the number of slots held at 108, three flags move to enrolled students and dropout recall falls from 0.507 to 0.493. Whether these seven flags are correct remains unknown.

Eligibility. Setting aside the 49 validation students who took no first-semester evaluation (36 of them dropouts) and re-ranking the remaining 495 with the frozen model, the top 20% (99 students) still contained only dropouts (precision 1.000), recall rose to 0.559 and ROC-AUC was 0.949 (Appendix Table 10). The flag is therefore not merely detecting students who had already stopped attending, although the 99 flagged students remain those with weak first-semester results.

Admission-time window. Without first-semester results, the validation ROC-AUC of TabICL fell from 0.952 to 0.874 and capacity recall from 0.507 to 0.469 (precision 0.926), with similar declines for all models; first-semester results thus carry most of the signal.

Without protected attributes. Refitting TabICL without gender, nationality, age, international status, displacement and special needs changed the validation ROC-AUC from 0.952 to 0.950, left recall and precision at 0.507 and 1.000, and retained 103 of the 108 flags (the five students swapped in and the five swapped out were all dropouts). The gender recall gap narrowed from 0.19 (0.404 vs. 0.597) to 0.10 (0.455 vs. 0.553).

5.6. Human-Study Quantitative Results

All 50 assigned participants consented and completed the study, and none were excluded, yielding 500 scored responses. Mean accuracy was 0.916 with profiles only and 0.900 with explanation charts (Table 8), a difference of -0.016 (95% interval -0.060 to 0.032; exploratory permutation $p = 0.61$; Hedges' $g = -0.19$). There is thus no evidence that the charts helped participants predict the flag. Participants with charts were, however, somewhat more confident ($+0.21$ points on a 1–5 scale, 0.02 to 0.42; $p = 0.055$), whereas self-reported time did not differ clearly.

Table 8. Participant-level results (25 per group; 95% bootstrap intervals).

	Profiles only	+ Charts	B − A
Accuracy	0.916	0.900	−0.016 [−0.060, 0.032]
Confidence (1–5)	4.14	4.35	+0.21 [0.02, 0.42]
Minutes (self-rep.)	33.4	30.2	−3.2 [−8.7, 2.0]

Accuracy was high in both groups, since most profiles were easy to judge: flagged cases usually showed no passed units. With charts, accuracy rose on flagged cases (0.968 to 0.992) and fell on unflagged ones (0.864 to 0.808), the fall coming entirely from one case. Participants described themselves as students (23), staff or advisers (23, a broad option) or other (4), with similar accuracy across roles. Nine of the ten cases had equal or higher accuracy with charts. The exception, case C01, was not flagged, yet its chart showed several factors pushing risk upwards; accuracy fell from 0.60 without the chart to 0.20 with it (Figure 7). The chart shows contributions relative to a typical student rather than the cohort cut-off and thus invites the inference that "more concern means flagged". Without C01, the difference becomes $+0.027$ (-0.013 to 0.076); this post-hoc check shows that the overall result depends on a single case but does not establish a benefit. In Group B, 21 of 25 participants passed both chart-reading checks.

5.7. Qualitative Findings

The written answers are summarised from a single coding pass (Section 4). When asked why the model decided as it did, 72% of chart-group answers reasoned from the chart, whereas the profile-only group cited academic results (46%) or several factors together (46%); personal or demographic

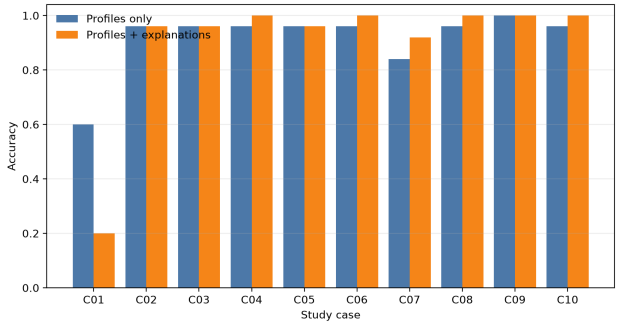


Figure 7. Accuracy by case and group; C01 had lower accuracy with charts.

reasons were almost never given (2 of 500). Suggested support was mostly academic (30%), followed by no action (21%, mostly for unflagged students), advising or monitoring (16%), financial or administrative help (15%) and first establishing what had happened (11%); wellbeing support was rare (2%). Overall, 76% of suggestions named a concrete next step (71% without charts and 82% with them), 91% could begin before the second semester and 2.8% asserted an unsupported cause. The chart group more often assigned the action to the institution (42% vs. 27%). Asked what else they would want to know, participants most often named academic engagement, such as attendance and current enrolment (40%), and personal circumstances, such as work or family (26%), none of which the dataset contains. Typical answers, paraphrased, were that the long first-semester bars pushed the estimate towards a flag; that the office should contact the student about unpaid fees, offer a payment plan and then arrange tutoring; and that it would help to know whether the student still attends and how many hours they work. The study was completed online, so these are written comments rather than think-aloud remarks. Answer depth varied: some answers were generic, and 21 support suggestions matched another participant's answer word for word.

6. Discussion and Limitations

6.1. Predictive Value Versus Interpretability

TabICL did not justify its complexity. Its advantage over XGBoost and the main-effects EBM lay within sampling noise on every metric, and EBM matched or slightly exceeded it at the operating point (recall 0.512 vs. 0.507; precision 1.000 vs. 0.991) while being transparent by construction. In line with Rudin (2019), the EBM (or logistic regression) would, on this evidence, be the preferable choice for any further development: it offers statistically indistinguishable performance, explanations that are the model itself rather than an approximation of it, no sampling instability and far cheaper inference (0.007 s versus 10.8 s to score the 545 test students on a CPU).

6.2. Interpreting the Near-Perfect Precision

The near-perfect precision does not indicate an excellent screening tool; it arises because the top 20% is filled with students whose records already show disengagement (Section 5.2). Such students are easy to rank, yet many may already be beyond the reach of a meeting and could be found with a transcript query; even among students who sat an evaluation, the flagged ones are those with weak results (Section 5.5). For an officer, the list adds an ordering, not new information. The more consequential group is the 105 unflagged validation dropouts, 93 of whom passed at least one unit. A simple rule could handle the obvious cases, reserving meetings for less visible students.

6.3. Implications of the Explanation Audit

The explanations were faithful under the deletion test and stable under the tested minimal grade perturbations. However, the exact list of top reasons often changes when the explanation is recomputed with a different seed or comparison group, so the bars should be read as an indication of what broadly mattered rather than as a precise ranking. Seed instability may partly reflect the one-permutation budget, the minimum that shap allows (a larger budget was not tested), whereas background instability reflects a substantive choice of which students count as "typical" and would not disappear with more computation. The local research interface (Appendix A.7) displays each student's stability alongside the reasons.

Fee status and recourse. Tuition fees up to date was the third most important group, and for 17 flagged students clearing a fee or debt record would remove the flag. In the model, the arrow runs from settled fees to a lower score, but this must not be read as "paying fees prevents dropout", since the causal arrow may point the other way: students already planning to leave, or in financial difficulty, may stop paying, making unpaid fees a symptom rather than a cause. A score change is not a causal effect (Karimi et al., 2021); an unpaid fee is better treated as a prompt for a conversation.

6.4. Interpreting the Human Study

The human study tested whether people could predict the model's flag, not whether explanations improve support decisions. For Group B the task partly became a chart-reading exercise, since adding up the bars largely reconstructs the flag (Section 5.4), and it was also easy without charts, as flagged profiles were visibly weak. The lack of an accuracy gain is therefore not proof that explanations lack value. Nevertheless, case C01 shows that bars displayed without the cohort cut-off can make an unflagged student appear flagged, and the higher confidence of the chart group, without any gain in accuracy, signals a risk of over-reliance.

6.5. Protected Attributes, Fairness and Deployment Boundaries

Whether gender, age or nationality should influence who receives a meeting is contested. Retaining them reflects real differences in base rates, and a model blind to them can still be unfair through proxies such as application route and course. On validation, however, removing them cost almost nothing (ROC-AUC 0.952 to 0.950), altered only 5 of 108 flags and narrowed the gender recall gap from 0.19 to 0.10; moreover, allocating outreach on the basis of who a student is, rather than how the student is performing, would be difficult to justify. The position adopted here is therefore to *audit* with protected attributes but not to *prioritise* with them. Subgroup results are descriptive, not a fairness certificate.

6.6. Threats to Validity

The evidence comes from one Portuguese institution without external or temporal validation, tests one foundation model (TabPFN was not compared), assumes rather than measures the 20% capacity, excludes enrolled students and rests on a single test set of 545 students whose intervals are wide relative to model differences. The full explanation audit followed the final test, using a refit that matches the frozen model, and its results depend on the SHAP budget and the 25-row background. The human study used volunteers, ten selected cases and an unsupervised online format; its comparisons are exploratory, and its coding is a single pass without measured agreement. None of the results identify causes of dropout or effects of interventions.

7. Conclusion

Regarding **RQ1**, TabICL ranked students well (ROC-AUC 0.957) but could not be distinguished from XGBoost, logistic regression or an interpretable EBM, so its additional complexity is not justified on this task. Regarding **RQ2**, its SHAP explanations were faithful but sensitive to the random seed and the comparison group, and for flagged students they largely restated weak first-semester results; fee or debt recourse existed for only 17 of 108 flags. Regarding **RQ3**, the study found no evidence that explanation charts improved participants' predictions of the flag (difference -0.016); participants with charts reported slightly higher confidence, while accuracy was lower with charts on one case. A transparent model is therefore the better candidate for further validation as a tool for *ordering* a contact list, provided an adviser reviews every flag alongside attendance, work and family, students who passed nothing are found by a simple check and the cohort cut-off is shown; its operational benefit remains untested, and it must never be used to deny support. Real use would first need validation on current cohorts and with support staff.

Acknowledgements

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Source Code and Data Availability

The code, frozen protocol and all aggregate results are available at https://github.com/TheInterpreters/Student-Dropout-Risk. The UCI dataset is public (CC BY 4.0); raw participant responses are not shared.

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APPENDIX

A. Supplementary Material

This appendix provides (A.1) team contributions, (A.2) reproducibility details, (A.3) additional predictive, faithfulness and subgroup results, (A.4) additional human-study results, (A.5) qualitative coding results, (A.6) the legal context and (A.7) the local research interface.

A.1. Team Contributions

Table 9. Main contributions of each team member.

Member	Contributions
Aye Khin Khin Hpone (Yolanda Lim)	Modelling pipeline, frozen protocol, one-time final test and model card; study cases, explanation charts and human-study analysis; sensitivity analyses; local research interface with what-if mode; integration, verification and reproducibility of all workstreams
Nguyen Liem Son (Lucas)	Seeded participant-assignment generator; explanation-chart and profile-card renderer with answer-leak check; unit tests and documentation for both
Witchayda Theppithuk (Noey)	Validation data and model-evaluation audits; full explanation audit on 544 validation students (faithfulness, stability, pair masking, surrogates and recourse); topic analyses, audit figures and results notebooks
Han Htoo Zaw (Han)	Google Forms specification, generator and Apps Script renderer; case-pack contract; response transformation; live form build, privacy settings and export checks
Pyae Sone Han (Harry)	Consent, instructions, debrief and retention arrangements; recruitment coordination; pilot; participant send log and data collection
All members	Recruitment of about ten participants each; report and presentation

A.2. Reproducibility and Provenance

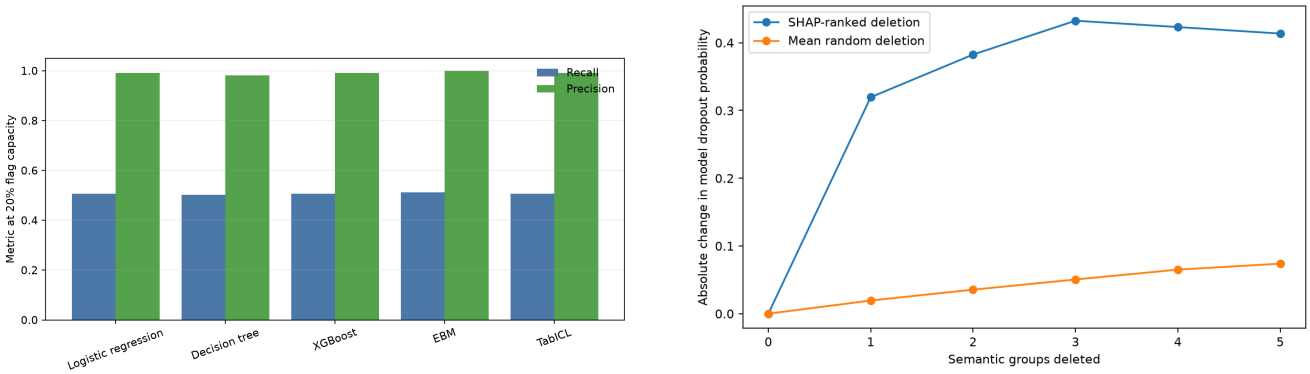
The split row identifiers, protocol, code snapshot and fitted model were hashed before the final test, and a lock file records that the test was run once (2 November 2026) and never repeated. The same final-test pipeline was first rehearsed on the validation split (saved separately as a dry run), and automated data checks on all 544 validation records found no non-finite values, impossible unit counts, out-of-range grades or duplicate profiles. All report numbers are regenerated from saved aggregate outputs with `python scripts/build_report_assets.py`, and `python scripts/verify_release_readiness.py` checks them against the saved evidence without fitting models or reading the test split. Raw survey responses, participant codes and the case-to-answer lookup are kept outside the repository, and only aggregate tables are shared. The repository separates source data, exploratory analysis, modelling, explanation evaluation, study materials and outputs; a pinned Python 3.12 environment and seed 42 support a fresh clone, `python scripts/launch_interface.py` builds the local interface, and a model card (intended use, data, final metrics, explanation evidence, subgroup results and limitations) is included in the project handoff document (`docs/HANDOFF.md`, Section 8). The test suite (98 unit tests) covers the modelling, explanation, study-integrity, interface and report-tooling code.

A.3. Additional Predictive, Faithfulness and Subgroup Results

With 109 slots and 213 test dropouts, recall cannot exceed 0.51 (Figure 8a), and removing groups in SHAP order moves the prediction far more than removing them in random order (Figure 8b). Table 11 reports final-test subgroups (left) and validation recall with and without protected attributes (right); gender follows the UCI coding (0 = female).

Global explanations. Three global views agree with the local SHAP ranking. Grouped permutation importance on 100 validation students (5 repeats) ranks first-semester progress far ahead (ROC-AUC loss 0.255), followed by tuition fees (0.038) and participation (0.025); every other group costs less than 0.005. The independently fitted EBM gives its largest mean contributions to first-semester pass rate, tuition fees, course, first-semester grade and scholarship status. ALE curves for admission grade, age at enrolment and credited units stay within ± 0.03 of the average prediction, so these inputs move the estimate little once first-semester results are known.

Pairwise masking. Masking pairs of each student's top five groups jointly rather than separately shows a mean absolute non-additive response of 0.036 probability units; the largest common pair is first-semester progress with participation (0.197 across 320 students). Adding pair terms reduced the local surrogate error from 0.073 to 0.022 for lasso (pair surrogates were better for 95% of students), so the five bars, read as separate effects, understate how these two first-semester groups act together.



(a) Final-test recall and precision at 20% capacity (b) Faithfulness on 544 validation students

Figure 8. Capacity-limited recall and precision (a) and deletion faithfulness (b).

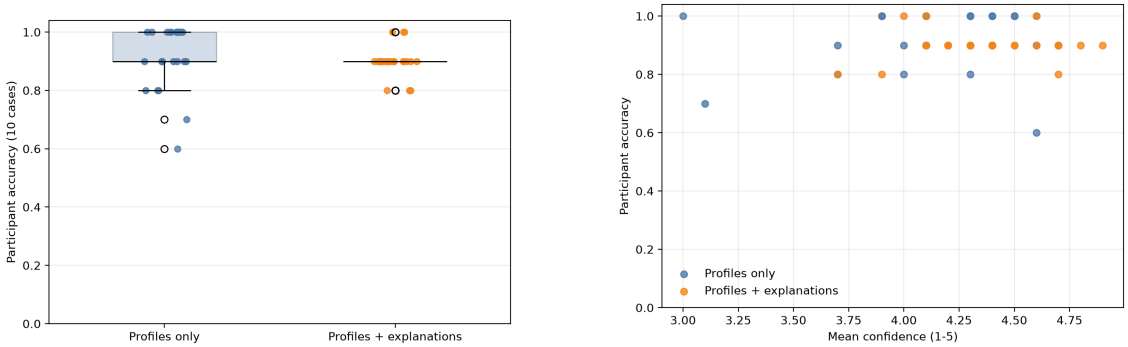
Table 10. Validation sensitivity analyses (exploratory; not used for model selection).

Setting	**ROC-AUC**	**Brier**	**Recall@20%**	**Precision@20%**
Primary (after first semester, 33 features)	0.952	0.071	0.507	1.000
Admission-time window (no first-semester results)	0.874	0.135	0.469	0.926
Without protected attributes (27 features, refit)	0.950	0.072	0.507	1.000
Enrolled share added as non-dropout (frozen model)	0.924	0.111	0.587	0.947
Enrolled share added as non-dropout (grouped refit)	0.929	0.085	0.606	0.977
Only students with $\geq$1 first-semester evaluation (frozen model, re-ranked)	0.949	–	0.559	1.000

Table 11. Descriptive subgroup diagnostics (– = suppressed, fewer than 30).

		Final test				**Validation recall**	
Attribute	**Group**	**n**	**Dropout rate**	**Flag rate**	**Recall**	**All features**	**No protected**
Gender	Female	335	0.299	0.122	0.410	0.404	0.455
	Male	210	0.538	0.324	0.593	0.597	0.553
Displaced	No	229	0.476	0.288	0.596	0.583	0.592
	Yes	316	0.329	0.136	0.414	0.409	0.398
Age at entry	$\leq$20	318	0.283	0.116	0.411	0.342	0.378
	21–25	93	0.430	0.194	0.450	0.565	0.500
	26–35	83	0.663	0.434	0.655	0.632	0.632
	36+	51	0.549	0.353	0.607	–	–

A.4. Additional Human-Study Results



(a) Accuracy by group (b) Confidence vs. accuracy

Figure 9. Participant-level accuracy and confidence.

Table 12. Case-level accuracy and the B − A difference without each case (95% intervals).

Case	Model flag	Acc. A	Acc. B	B − A without this case
C01	No	0.60	0.20	+0.027 [−0.013, 0.076]
C02	No	0.96	0.96	−0.018 [−0.062, 0.036]
C03	No	0.96	0.96	−0.018 [−0.062, 0.031]
C04	No	0.96	1.00	−0.022 [−0.067, 0.031]
C05	Yes	0.96	0.96	−0.018 [−0.062, 0.027]
C06	Yes	0.96	1.00	−0.022 [−0.067, 0.027]
C07	No	0.84	0.92	−0.027 [−0.067, 0.013]
C08	Yes	0.96	1.00	−0.022 [−0.067, 0.027]
C09	Yes	1.00	1.00	−0.018 [−0.067, 0.036]
C10	Yes	0.96	1.00	−0.022 [−0.071, 0.031]

A.5. Qualitative Coding Results

Table 13. Primary themes of the 1,500 written answers (single coding pass).

Question	Theme	%	Question	Theme	%
Why flagged	Explanation chart	36.0	Suggested support	Academic support	29.8
	Academic evidence	31.0		No action	21.0
	Several factors	28.0		Advising or monitoring	16.0
	Financial or admin.	3.0		Financial or admin. support	15.2
	Other	2.0		Verify information first	11.4
			Missing information	Academic engagement	39.8
				Personal circumstances	25.8
				Course or institution	12.0
				Finances	9.8
				Nothing / unclear	12.4

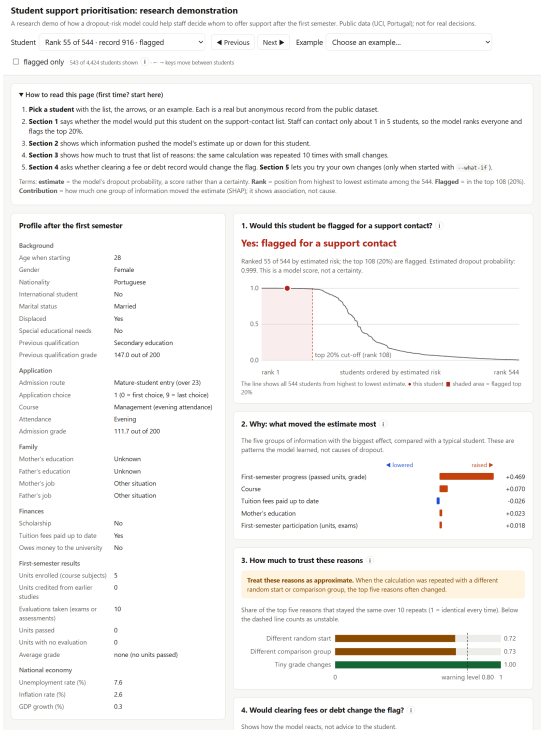
A.6. Legal Context

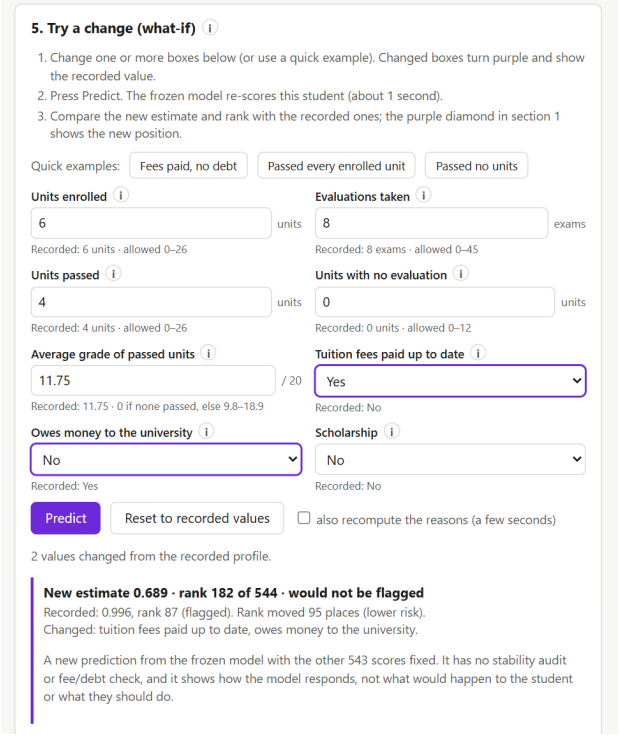
No formal legal review was carried out; this section indicates which data-protection rules would be relevant if the approach were taken further. The UCI dataset is anonymised and public, so the modelling itself processes no personal data. The participant responses are pseudonymous rather than anonymous while the case-linkage files exist, and would count as personal data under both the EU General Data Protection Regulation (GDPR) (European Parliament and Council of the European Union, 2016) and Thailand's Personal Data Protection Act (PDPA) (Kingdom of Thailand, 2019). Both laws rest on a lawful basis such as consent, purpose limitation, data minimisation and security, which the study addressed through informed consent, no names, random codes, restricted access and use for this course project only.

A deployed outreach system would raise further obligations. Real student records are personal data, and attributes such as health or special needs may be sensitive data requiring stronger safeguards. GDPR Article 22 restricts decisions based solely on automated processing that significantly affect a person (European Parliament and Council of the European Union, 2016; Wachter et al., 2017); the outreach rule in this report keeps an adviser responsible for every contact and never withholds support, but an institution would still need to inform students, assess the data-protection impact and confirm the rules of its own jurisdiction before any use.

A.7. Local Research Interface

To illustrate how the outputs could reach a support officer, a self-contained local web page was built for the 543 validation students not used in the study (Figure 10). For each student, it shows whether the student would be flagged and their rank on the cohort risk curve, the five largest grouped contributions, the stability of that explanation, the fee/debt recourse check (labelled as model behaviour rather than advice) and the profile. An optional local what-if mode re-scores an edited copy of a student with the frozen model, allowing changes to first-semester results and fee, debt or scholarship status only. It keeps the other 543 scores fixed, rejects impossible combinations (for example, more units passed than enrolled) and reproduced all 104 saved fee/debt scenarios (largest probability difference 4.1×10^{-6}, identical ranks). Figure 10(b) applies it to a flagged student: clearing the unpaid fees and the debt lowers the estimate from 0.996 to 0.689 and the rank from 87 to 182, so the student would no longer be flagged. Such results describe how the model responds, not what would happen to the student. The interface was built after the study, uses validation data only, runs solely on the local machine and was not tested with users; it is therefore a demonstration rather than evidence.





(a) Saved view: flag, rank, reasons and their stability (b) What-if mode: fees paid and debt cleared

Figure 10. Local research interface (validation students only): (a) saved view for student 916; (b) what-if mode for student 174.